Developing and Operationalising Natural Language Processing solutions for Child Safeguarding: A Critical Review of Technical Possibilities and Organisational Realities in UK Policing – TITLE TO BE CONFIRMED!!!

Theoretical Perspectives of Advanced Professional Practice - MOD006047

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Abstract

Purpose: Lorem impsum...

Method: Lorem impsum...

Findings: Lorem impsum...

Conclusion: Lorem impsum...

Contents

Acronyms

AB AdaBoost. 20

ACL Association for Computational Linguistics. 32

ACM Association for Computing Machinery. 32

AI Artificial Intelligence. 16, 23–26, 32, 33

ANT Actor Network Theory. 23

BERT Bidirectional Encoder Representations from Transformers. 20

BiLSTM Bidirectional Long Short-Term Memory. 20

BLSTM Bidirectional Long Short-Term Memory. 20

BPNN Back-Propagation Neural Network. 20

CADS Corpus Assisted Discourse Studies. 17

CEOP Child Exploitation and Online Protection. 37

CNN Convolutional Neural Network. 20

CPS Crown Prosecution Service. 23

CSA Child Sexual Abuse. 7, 9, 12–15, 21, 37

CSEW Crime Survey for England and Wales. 12, 13

DPC Data Processing Contract. 26, 27

DPIA Data Protection Impact Assessment. 26, 27

DT Decision Tree. 20

E2EE End-to-End Encryption. 19

EU European Union. 33

GB Gradient Boosting. 20

GRU Gated Recurrent Unit. 20

HMICFRS His Majesty's Inspectorate of Constabulary and Fire & Rescue Services. 23

ICO Information Commissioner's Office. 37

IEEE Institute of Electrical and Electronics Engineers. 32

KNN K-Nearest Neighbours. 20

LCT Luring Communication Theory. 17

LLM Large Language Model. 16

LogR Logistic Regression. 20

LR Logistic Regression. 20

LSTM Long Short-Term Memory. 20

LWIC Linguistic Inquiry and Word Count. 17

ML Machine Learning. 16, 20

MLP Multilayer Perceptron. 20

NB Naïve Bayes. 20

NIST National Institute of Standards and Technology. 37

NLP Natural Language Processing. 7–10, 15, 16, 24, 31, 32, 37, 38

NPCC National Police Chiefs' Council. 37

NSPCC National Society for the Prevention of Cruelty to Children. 12

OGDM Online Grooming Discourse Model. 13–15, 17

PJ Perverted Justice. 15, 16, 18, 20

PKMS Personal Knowledge Management System. 10

PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analyses. 10

RF Random Forest. 20

Ridge Ridge Regression. 20

RNN Recurrent Neural Network. 20

SGM Sexual Grooming Model. 13–15, 17

SRMSG Self-Regulation Model of Sexual Grooming. 14, 15

SSRN Social Science Research Network. 33

SVM Support Vector Machine. 20

SVN Support Vector Network. 20

TAM Technology Acceptance Model. 23

TLP Transformer Led Policing. 23, 24

UK United Kingdom. 12, 25, 26

UTAUT Unified Theory of Acceptance and Use of Technology. 23

VKPP Vulnerability Knowledge & Practice Portfolio. 12

ZK Zettelkasten. 10

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Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four thematic research areas to develop a conceptual framework capable of addressing the overarching research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

The four thematic research areas are as follows:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
2. Which NLP methods can beused to identify these precursors?
3. What implementation barriers exist within policing that may hinder NLP adoption?
4. What are the ethical and statutory challenges in implementing an NLP solution for safeguarding?

These questions provide the scaffoldinge structure of the literature review described in section 1.2.

1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. Each track corresponds directly to one of the four sub-questions introduced in section 1.1, providing the evidential foundation from which each question will be addressed.

The insights gathered across these four tracks are subsequently then synthesised using integrative thematic analysis, identifying points of convergence and tension between them and resulting in a conceptual framework which grounds the evolving research questions presented in chapter 7.

Chapter 2

Literature Search Strategy and Methodology

2.1 Review Methodology and Rationale

This literature review adopts a systematic integrative approach. Snyder (2019) documents a number of review types, noting that an integrative review seeks to combine multiple perspectives, synthesise those views, and to develop a new framework or perspective.

The interdisciplinary nature of this research, noted in section 1.2, necessitates a review methodology that can accommodate multiple perspectives and evidence types. An integrative design as defined by Torraco (2016) and Torraco (2005) allows for a review and synthesis across domains, building a conceptual framework which seeks to unify the disparate domains into a framework for operationalising and implementing an NLP solution into a safeguarding CSA context.

A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identify that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demanded a robust approach to literature and note management. The Zettelkasten (ZK) method, described by Basu (2020) and Reck (2023) as a process of building a Personal Knowledge Management System (PKMS), was adopted to manage sources, create thematic links between them, and support synthesis across the four research tracks. Following Reck (2023), a supporting software stack was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note-taking and synthesis, and LaTeX for writing and formatting. The application of this approach is illustrated in figure 2.1.

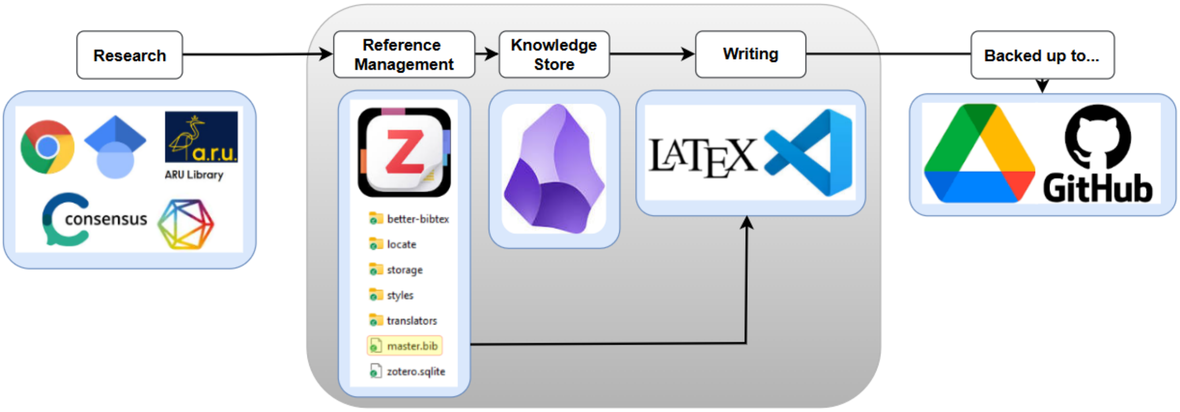

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

The search strategy for this review, including database selection, keywords, and inclusion and exclusion criteria for each of the four research tracks, is presented in full in appendix A.

2.4 Keywords and search terms

Keywords and Boolean search strings for each research track, developed in accordance with Chigbu et al. (2023), are defined in appendix B.

2.5 Inclusion and exclusion criteria

Appendices C and D present the inclusion and exclusion criteria for each of the four research tracks. The year date of 2018 is presented as this is largely considered to be the breakout year for the development of NLP transformer based models, (Khanna, 2024), a technical architecture discussed in chapter 4.

2.6 Results and filtering process

A Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) diagram is presented in figure 2.2 which summarises the selection process following the implementation of the search strategy.

From an initial return of 3,150 records, a three-stage screening process reduced the corpus to 100 papers for review, supplemented by 22 purposefully selected seminal works.

2.7 Reflections on the search and filtering process

Reflecting on this process identifies limitations in search precision. The Boolean strings presented in table B.1 were insufficiently refined, returning a volume of literature that the inclusion and exclusion criteria were not robust enough to filter efficiently, requiring a degree of subjective judgement that a tighter search strategy may have reduced.

Whilst the final selection of papers used in this review was informed by a structured and repeatable process, this experience highlights the value of iterative Boolean refinement prior to full database submission. Future searches within this doctoral programme will incorporate pilot searches to test string precision before full execution, reducing reliance on post-hoc subjective filtering.

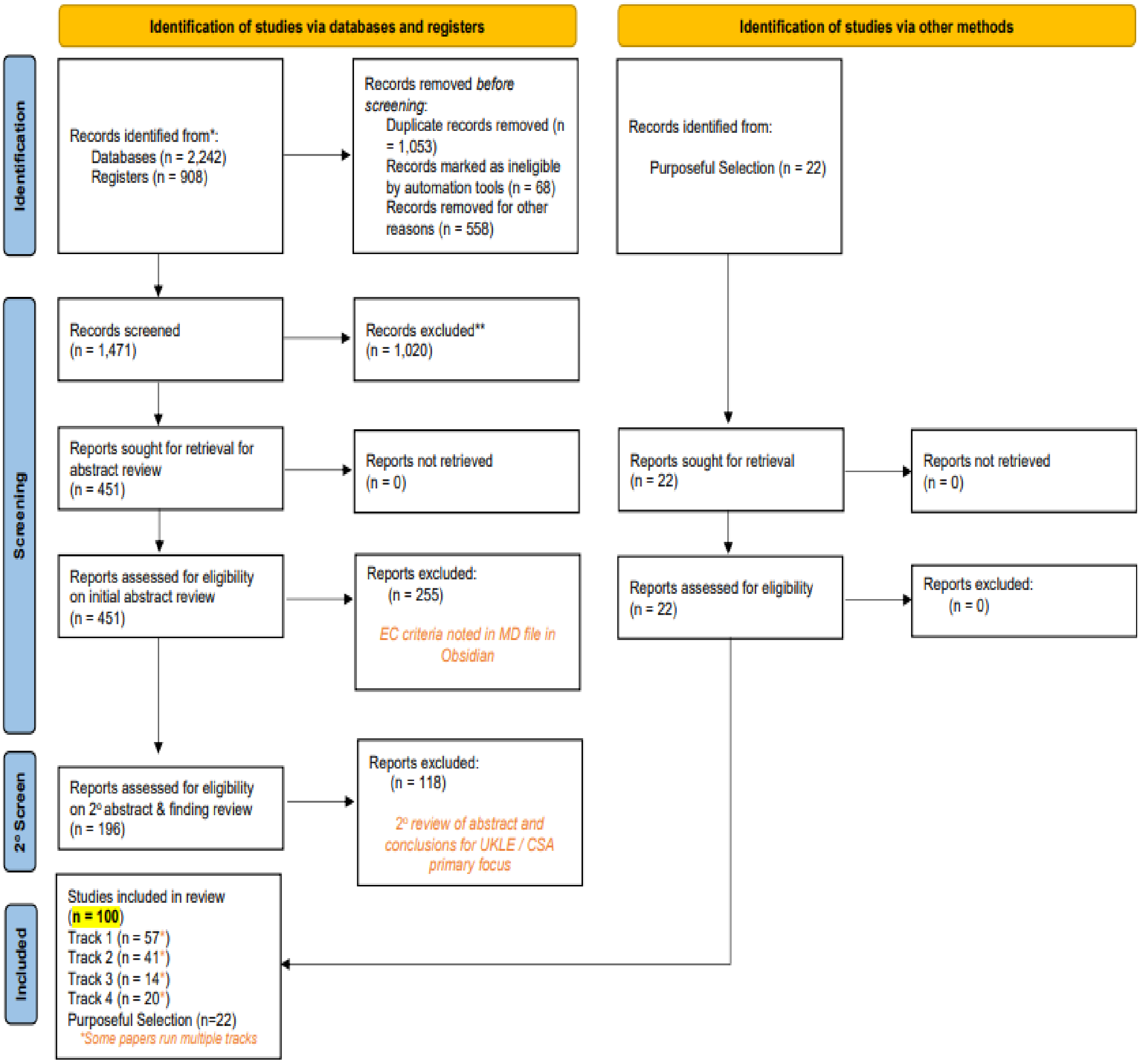

Figure 2.2: PRISMA diagram summarising the literature search and filtering process

Chapter 3

The Landscape of Child Sexual Exploitation and Policing's Response

3.1 The scale of CSA reporting

The measurement of crime statistics in the United Kingdom (UK) is complex, with two primary sources of data providing different perspectives on crime (Ariel and Bland, 2019). The Crime Survey for England and Wales (CSEW) acts as a household survey capturing participants' self-reported experiences of crime, whilst police recorded crime data captures the volume of offences formally reported to and recorded by police forces.

Neither dataset can be considered a definitive measure of crime, and this limitation is thrown into sharp relief when examining CSA.

The Vulnerability Knowledge & Practice Portfolio (VKPP) reported 122,768 CSA offences in 2024 (VKPP, 2025) taken from police recorded crime data, whilst the 2024 CSEW estimates that 1 million people had experienced sexual abuse in the previous year, (Office for National Statistics, 2024).

It is important to note that these two datasets are not directly comparable. The VKPP CSA figure represents criminal offences reported to police forces within a single calendar year. The CSEW figure is a cumulative prevalence estimate drawn from a cross-sectional household survey of adults aged 16 and over recalling experiences.

These discrepancies in reporting the scale of CSA offending highlights the challenges in measuring the scale of the issue and the demand it places on policing. The VKPP figure represents the formal demand on policing resources, while the CSEW figure suggests a much larger underlying prevalence of CSA that may not be fully captured by police recorded crime data. Additionally, Whilst the described reporting of CSA focusses on physical offending, Finkelhor et al. (2024) suggests that online sexual abuse and communication, when taken into consideration, would markedly increase the prevalence of CSA.

This is a view supported by Wefers et al. (2024) who leans into figures provided by the National Society for the Prevention of Cruelty to Children (NSPCC). Whilst the source of Wefers et al. (2024)'s NSPCC data is no longer accessible, more contempory reporting by the NSPCC suggests an 89% increase in sexual communication offences between 2018 to 2024, (NSPCC, 2024).

Starkly, these NSPCC figures relate to only 7,000 reports captured by NSPCC Freedom of Information requests to police forces, (NSPCC, 2024). The research of Finkelhor et al. (2024) and Wefers et al. (2024), when aligned to this low number, suggests that the scale of CSA associated to online sexual abuse and communication is not being fully captured by police recorded crime data, and that there may be a significant gap between the prevalence of online sexual communications and the demand it places on policing resources.

3.2 The demand on law enforcement

Both academic and grey literature identify that the demand on policing resources from CSA is increasing, with the gap between demand and capacity widening.

Jay et al. (2022) suggests that the scale of known CSA demand may never be met by law enforcement, identifying that digital forensic delays and a constantly shifting set of policing priorities both impact on law enforcement's ability to tackle the scale of the issue. Casey (2025) adds to this summation, noting complexities around fragmented policing systems and a lack of intelligence sharing further exacerbate the potential to tackle demand.

Beyond these structural blockers, the challenges operate on two distinct dimensions. Li (2020) identifies the first as one of volume and process: the manual review of suspect communications is lengthy and resource-intensive, typically occurring after a contact offence has already taken place. Winters et al. (2022) identify a second, more fundamental dimension: that law enforcement does not fully understand the linguistic and behavioural complexities inherent in grooming communication itself, limiting the capacity for proactive identification even where resource permits.

Together with the prevalence gap identified by Finkelhor et al. (2024) and the CSEW, these dimensions suggest that policing is largely confined to reacting to known demand rather than proactively safeguarding children. This has direct implications for the potential of technological intervention, and raises the question of whether the linguistic complexities of grooming communication are computationally tractable — a question the following sections begin to address.

3.3 The grooming to contact offending pathway

The academic literature which describes the process of grooming a child before the commission of a sexual offence is expansive and complex. Three contemporary grooming models have been identified from the literature;

Winters and Jeglic (2017) provide a five stage Sexual Grooming Model (SGM). This model has been subject to review and validation, (Winters et al., 2020) and has led towards the consideration by Winters et al. (2022) of a universal definition of grooming. The SGM however focusses exclusively on in person grooming pathways, and does not offer insight into how grooming may occur in an online or digital context.

Lorenzo-Dus et al. (2016) provides a staged model of grooming focussing on the online context only. This model, entitled the Online Grooming Discourse Model (OGDM), has been expanded over time using growing corpora of conversational data, (Lorenzo-Dus and Izura, 2017; Lorenzo-Dus and Kinzel, 2019; Lorenzo-Dus et al., 2020) and has been considered in operational law enforcement settings, Swansea University (2023).

Whilst the models presented by Winters and Jeglic (2017) and Lorenzo-Dus et al. (2016) are practical in nature, Elliott (2017) presents a more theoretical model of grooming, describing the process as a self-regulatory process which is influenced by both an offender's psychological make up in addition to an ongoing external feedback loop affecting external behaviours. This model is less practical in nature but provides a structured theoretical framework through which to understand the grooming process.

A summary table of these three models is presented in table 3.1.

Table 3.1: Comparison of Three Models of Sexual Grooming

Feature	**Sexual Grooming Model (SGM)**	**Online Grooming Discourse Model (OGDM)**	**Self-Regulation Model of Sexual Grooming (SRMSG)**
Source	Winters (2017, 2020, 2022)	Lorenzo-Dus (2016, 2017, 2020)	Elliott (2017)
Context	Focus on in-person (offline) grooming.	Focus on online (computer-mediated) environments.	Theoretical universal model.
Stages	Five stages: (1) victim selection, (2) gaining access and isolation, (3) trust development, (4) desensitisation, and (5) post-abuse maintenance.	Three phases: (1) Access, (2) Entrapment, and (3) Approach.	Two phases: (1) Potentiality and (2) Disclosure.
Basis of Model	Synthesised grooming literature to identify observable and measurable behaviours, latterly validated.	Founded on Computer-Mediated Discourse Analysis, Speech Act Theory, and Relational Work.	A synthesis of previous models grounded in Control Theory.
Strengths	Intuitive and understandable to practitioners. Empirically validated via expert review.	Notes that language can be multi functional in building trust and also enabling desensitisation (Speech Act Theory). Also captures the ability of a groomer to scale activities to multiple potential victims.	Synthesis of multiple models grounded in well established control theory structure.
Limitations	Focusses solely on in-person (offline) grooming, thereby limiting generalisability to online contexts.	Focusses solely on the online context, initially with limited data, much of which was involving decoys as opposed to real-world scenarios.	Entirely theoretical model which is potentially too abstract for practical application.

3.4 Synthesising the existing grooming models

The models presented in table 3.1 all identify that the grooming process is patterned, staged and, in principle, identifiable (albeit with mixed success, (Winters and Jeglic, 2016, 2017)).

Each of the models offers are supported by degrees of rigor, with the SGM being empirically validated by experts in the CSA field, the OGDM leaning into the computational linguistics and grounding itself in Speech Act Theory, Lorenzo-Dus et al. (2016); Lorenzo-Dus and Izura (2017), and the Self-Regulation Model of Sexual Grooming (SRMSG), presenting as a theoretical model grounded in Control Theory, Elliott (2017).

Whilst these models advance scholarly understanding of the grooming process, they are incommensurable. The SGM concerns itself entirely with in-person grooming and, whilst validated by experts, is based on fictional "vignettes" and not on real world data. The OGDM focusses on the online grooming context only, using chat data captured from the well known open source data set presented by the Perverted Justice (PJ) Foundation; http://www.pjfi.org/. This data set, collected between 2004 and 2016 presents adult-decoy interactions with online offenders, potentially missing the nuance and language that may be present between an adult groomer and a child victim. The SRMSG is a theoretical model which synthesises previous models and is grounded in Control Theory, but it does not offer practical insights into the grooming process and is not based on real world data.

The difficulty in synthesising these models can be seen in the effort by Winters et al. (2022) who attempt to produce a universal definition of sexual grooming. They produce a definition based on a systematic review of thirteen preceding definitions, however their concluded definition text is grounded in their SGM and so is inherently biased towards in-person grooming.

None of the presented models leverage real operational data taken from a law enforcement context, and whilst all of the models define a staged process of grooming, none of them are able to apply real-world insight into how contact sexual offences manifest from the grooming conversation.

3.5 The case for technological intervention - Identifying the research opportunity

The work of Winters and Jeglic (2016, 2017) identify the difficulties inherent in human identification of grooming content. These observations are compounded by Li (2020) who notes the lengthy periods that are spent by law enforcement in reviewing suspect chat logs, often in circumstances after a contact offence as occured.

Together, these challenges present a compelling case for technological intervention. Chapter 4 evaluates the technical landscape of NLP in the CSA context, assessing the current landscape of computational approaches to grooming detection.

Chapter 4

NLP and Computational Approaches to Grooming Detection

4.1 Natural language processing. A technical baseline

NLP is defined by Jose Gonzalez-Gomez et al. (2024) as a subset of AI which *works with text processing capabilities to understand texts like humans do*.

Amongst the many use cases which can be applied to NLP technologies, the concept of text classification is relevance to this research. Text classification is an established Machine Learning (ML) methodology whereby algorithms are trained on labelled data to categorise text into predefined classes. The trained algorithm can then be applied to new, unseen text to predict the appropriate category based on learned patterns and features, (Belcic, 2024; Jose Gonzalez-Gomez et al., 2024).

The technical landscape of NLP has been transformed by the development of transformer architecture, (Vaswani et al., 2017), which enables a parallel computational processing approach as oppsoed to a sequential one,capturing contextual relationships between words and providing the foundation for Large Language Model (LLM) technologies capable of *understanding* language at scale, (Khanna, 2024; Raiaan et al., 2024).

4.2 Classification in the grooming context

As early as 2012, computer scientists were exploring the application of ML and NLP techniques to the detection of grooming content in online conversations, (Inches and Crestani, 2012). Since then, there has been a growing body of technical research in this area, with a predominent focus on classifying text content as either being indicative of grooming or not grooming.

Leiva-Bianchi et al. (2025) provides a recent systematic review of the published literature which tackles the grooming classification problem identifying relevant ML algorithms used, the datasets which fed that classification process and the effectiveness of those algorithms. A summary of of Leiva-Bianchi et al. (2025)'s work is presented in table 4.1. It is immeciately apparent that the PJ dataset, and derivatives of it, noted as PAN12 or PAN13, are the predominant data source for this research, with a significant number of studies using these datasets to train and test their algorithms. The PJ dataset is noted in section 3.4 as being a dataset of adult-decoy interactions with online offenders. Further considerations around this dataset are explored in section 4.3.

The predominant goal across the studies shown in table 4.1 was to provide accurate classification as to whether a given message or conversation could be classified as being indicative of grooming or not. It is notable from the commentary provided by Leiva-Bianchi et al. (2025) that whilst the studies rightly concern themselves with the accuracy of classification, there is a lack of focus in grounding any accuracy metric against any theoretical grooming models such as those described in section 3.4.

Some studies not captured by Leiva-Bianchi et al. (2025) do seek to provide a criminological or psychological grounding to their classification approaches;

Hamm and McKeever (2025) ground their grooming detection methodology in Social Exchange Theory, noting that the grooming process is one of social transactional exchange between the offender and the victim. Hamm and McKeever (2025) use this theoretical grounding to develop a model in which positive and negative sentiment identified from message strings are used as part of the classification process.

An et al. (2025) use Luring Communication Theory (LCT) to frame their classification approach. LCT is a grooming framework proposed by Olson et al. (2007) which ientifies a five part cyclical grooming process, shown in figure 4.1. An et al. (2025) use LCT as the labelling basis for their classification training process.

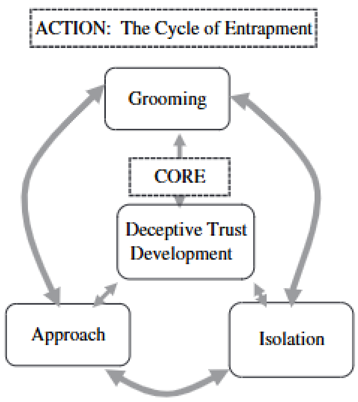

Figure 4.1: Luring Communication Theory, taken from Olson et al. (2007)

Whilst the papers considered in table 4.1 and noted above focus predominantly on the technical aspects of grooming detection, Lorenzo-Dus and Kinzel (2019) and Lorenzo-Dus et al. (2020) provide a more linguistically grounded approach. Whilst neither paper provides a technical classification model, both employ technical analysis of linguistic features using methodologies such as Corpus Assisted Discourse Studies (CADS) and Linguistic Inquiry and Word Count (LWIC), with Lorenzo-Dus et al. (2016); Lorenzo-Dus and Izura (2017); Lorenzo-Dus et al. (2020) grounding their analysis firmly in established linguistic theory.

Cumulatively however, the literature predominantly focuses on binary language classification (grooming vs. non-grooming), often without clear grounding in a criminological or psychological grooming model.

Treating grooming detection as a binary classification problem reduces the grooming process to a discrete state rather than recognising it as the dynamic, escalating process described in the literature. Section 3.3 describes the SGM and OGDM models, both of which note a sequential progression through distinct grooming stages. This multi-stage process is similarly reflected in Olson et al. (2007)'s LCT model, shown in figure 4.1.

The escalatory and sequential nature of the grooming process suggests that linguistic markers indicative of escalation toward contact offending may be identifiable within conversational data. This identification of this threshold, the point at which an online grooming offence transitions toward physical contact, remains underexplored in the existing literature, representing a significant gap with direct relevance to the demand concerns across operational proactive safeguarding, noted in section 3.2.

4.3 The data provenance challenge

Table 4.1 clearly shows the predominant use of the PJ dataset across research in this area.

The PJ dataset, and derivatives of it (PAN12, PAN13), are the predominant data source for the technical research in this area. The PJ dataset is a collection of adult-decoy interactions with online offenders. Derivatives of this data set appended non-offensive content to the decoy data to create a more comprehensive dataset. Despite its widespread use, the dataset has significant limitations which are summarised as follows:

- The use of adult decoys is described as lacking authenticity and potentially being unrepresentative of real-world grooming conversations, An et al. (2025); Bihani et al. (2025); Evans and Lorenzo-Dus (2025).
- The dataset was collected between 2004 and 2016. It is noted that the decade that has followed has seen new communication patterns and platforms emerge, potentially leaving the language and communication styles in the PJ dataset as not reflective of current grooming conversations, Agarwal et al. (2022); Xiao (2025); Lorenzo-Dus et al. (2020).
- The original PJ dataset is limited in size, containing only a small number of grooming conversations. This lack of diversity in the dataset may limit the effectiveness of any models trained on it, An et al. (2025); Inches and Crestani (2012).
- There has been an over reliance on the PJ dataset across the research community, further limiting the generalisability of any results, An et al. (2025); Leiva-Bianchi et al. (2025).

Whilst algorithmic models can be trained on synthetic data, the lack of real world data is a significant barrier to the development of an operationally deployable model. The language and communication styles in real world grooming conversations may differ significantly from those in the PJ dataset, and without access to real world data, it is difficult to conceive that any model not trained on real world data can effectively identify grooming content in an operational context.

4.4 The operational gap

Whilst section 4.3 identifies the data challenge in classifying grooming content, An et al. (2025) identifies that efforts to technically classify chat content as "grooming" or "not grooming" have primarily been considered from a post-hoc perspective. This "after the event" classification mirrors the operational reality noted in section 3.2 where law enforcement are often in a position of having to review large volumes of chat data after a contact offence has occurred. The opportunity for proactive safeguarding, to identify grooming content before it escalates toward contact offending, remains underexplored in the existing literature. An et al. (2025) makes the case for attempting to the identify the optimal intervention point in the grooming process, however, as with the majority of the existing literature, their work is based on the PJ dataset and so is limited by the data provenance challenge noted in section 4.3.

Whilst many of the classification efforts claim success in proving concept in identifying grooming content, those that progress to a potential real world use case lean towards considering detection into social network chat groups or capturing live feeds in online conversations. S. E. Abdelhamid and M. S. Elhussiny (2026) discuss the delivery of a technical solution entitled **SafeGuard** which theorises the delivery of a grooming detection model which coould be used in the home or in schools, however this does not progress into realisation. Agarwal et al. (2022) and Xiao (2025) both contextualise their classification efforts within the realms of instant messaging platfroms, but neither offer real world deployment insight. One of the possible reasons for this is noted by Street et al. (2025) who notes the prevalence of End-to-End Encryption (E2EE) across modern messaging platfroms making the capture of live data for real world deployment a significant challenge.

For the law enforcement context specifically, the operational gap remains significant. Multiple studies identify the potential for grooming classification to have a positive impact on safeguarding practice across law enforcement, (Inches and Crestani, 2012; Agarwal et al., 2022; Hamm and McKeever, 2025) but there appears to be a notable absence of research which considers the practicalities of deploying a potential technical solution to grooming classification within a law enforcement context. This challenge is discussed in chapter 5.

Table 4.1: Summary of reviewed studies - Acronyms and abbreviations defined in the table header

Abbreviations: AB = AdaBoost; BERT = Bidirectional Encoder Representations from Transformers; BiLSTM = Bidirectional Long Short-Term Memory; BLSTM = Bidirectional Long Short-Term Memory; BPNN = Back-Propagation Neural Network; CNN = Convolutional Neural Network; DT = Decision Tree; GB = Gradient Boosting; GRU = Gated Recurrent Unit; KNN = K-Nearest Neighbours; LogR = Logistic Regression; LR = Logistic Regression; LSTM = Long Short-Term Memory; MLP = Multilayer Perceptron; NB = Naïve Bayes; PJ = Perverted Justice; RF = Random Forest; Ridge = Ridge Regression; RNN = Recurrent Neural Network; SVM = Support Vector Machine; SVN = Support Vector Network.

Authors and Year	**ML Algorithm**	**Dataset**	**Metrics**	**Quality**
Agarwal et al. 2022	MLP	**PAN12**	P, R, F1, F05	High
Amuchi et al. 2012	NB, SVM	Private Chat	ACC	Medium
Anderson et al. 2019	AB, LogR, NB, RF, SVM	**PAN13**, **PJ**	ACC	High
Andleeb et al. 2019	NB, SVM	MySpace, **PJ**	ACC, P, F1	High
Ashcroft et al. 2015	AB	**PAN12**, **PAN13**	ACC	High
Bogdanova et al. 2014	SVM	CYBERSEX, NPS, **PJ**	ACC	Medium
Borj et al. 2019	NB, RF, SVM	**PAN12**	ACC, P, R, F1	High
Borj et al. 2020	NB, RF, SVM	**PAN12**	ACC, P, R, F1, F05, F2	High
Borj et al. 2021	GB	**PAN12**	ACC, F1	High
Borj et al. 2023	SVM, RF, NB	**PAN12**	ACC, P, R, F1, F05	High
Bours et al. 2019	LogR, MLP, NB, Ridge, SVM	**PAN12**	P, R, F1, F05, F2	High
Cano et al. 2014	SVM	**PJ**	P, R, F1	High
Cardei et al. 2017	RF, SVM	**PAN12**	P, R, F05	High
Cheong et al. 2015	DT, KNN, LogR, MLP, NB, RF, SVM	MovieStarPlanet	ACC, P, R, F1, F05	High
Dhouioui et al. 2016	SVM	Private Chat	F1	Medium
Ebrahimi et al. 2016	CNN, MLP, SVM	**PAN12**	P, R, F1	High
Ebrahimi, Suen et al. 2016	NB, SVM	**PAN12**	ACC, P, R, F1	High
Faraz et al. 2024	GB, KNN, SVM	**PAN12**	ACC, P, R, F1, F2, F05	High
Fauzi et al. 2020	DT, KNN, LogR, MLP, NB, RF, SVM	**PAN12**	ACC, P, R, F1, F05	High
Fauzi et al. 2023	SVM	**PAN12**	ACC, P, R, F1, F05	High
Gunawan et al. 2016	KNN, SVM	LITEROKA, **PJ**	ACC	Medium
Hamzah et al. 2021	BiLSTM, GB, GRU	Chat n/s, **PAN12**	ACC	Medium
Isaza et al. 2022	CNN	**PAN12**	ACC, P, R, F1, S	Medium
Kim et al. 2020	RNN	**PAN12**	P, R, F1, F05	Medium
Kirupalini et al. 2021	AB, KNN, NB, RF, SVM	FUGLY, IRC, **PJ**	ACC, P, R, F1	High
Melleby 2023	BERT	**PAN12**, AIBA AS	P, R, F1	High
Michalopoulos et al. 2014	KNN, NB, SVM	**PJ**	ACC	Medium
Milon et al. 2022	MLP	IRC, **PJ**	F1	High
Misra et al. 2019	CNN	**PAN12**, **PJ**	F1	Medium
Mohammed et al. 2024	SVM, NB, RF	**PAN12**	ACC, P, R, F1	Medium
Morris et al. 2013	SVM	**PAN12**	P, R, F1	High
Munoz et al. 2021	CNN	Chat n/s, **PAN12**	ACC, R, F1	High
Ngejane et al. 2021	BLSTM, GB, LogR, MLP	**PAN12**	ACC, P, R, F1	High
Nyrem et al. 2023	NB, DT, KNN, LogR, MLP, RF	**PAN12**	ACC, P, R, F1	High
Pandey et al. 2012	SVM	Omegle, **PJ**	ACC	Medium
Parapar et al. 2014	SVM	**PAN12**	P, R, F1	High
Peersman et al. 2012	SVM	**PAN12**	P, R, F1	High
Pranoto et al. 2015	LogR	**PJ**	ACC, P, R	Medium
Preuß et al. 2021	MLP	**PAN12**	P, R, F05, F2	High
Ringenberg et al. 2019	CNN	**PJ**	NA	Medium
Suhartono et al. 2019	BPNN	**PJ**	ACC	High
Vartapetiance et al. 2014	NB, SVM, DT	**PAN12**	P, R, F1	High
Vogt et al. 2021	BERT	CHATCODER, **PAN12**	P, R, F1	High
Waezi et al. 2024	SVN, RNN, LSTM, GRU	**PAN12**	F2, F05	High
Zambrano et al. 2019	LR	**PJ**	ACC	Medium
Zuo et al. 2018	AB, LogR, NB, RF	Chat n/s, **PAN13**	ACC	Medium
Zuo et al. 2019	BPNN	Chat n/s, **PAN13**	ACC	Medium

Chapter 5

Technology adoption and implementation in Policing

5.1 UK Policing: Cultural fragmentation and an innovation deficit

Academic literature and independent governmental reviews identify three recurring challenges to technology adoption across UK policing: structural fragmentation, cultural resistance to change, and organisational barriers. Evidence of these challenges is presented in table 5.1.

Section 3.5 notes commentary by Winters and Jeglic (2016, 2017) and Li (2020) who all note the need for technology to assist in the CSA investiative and safeguarding process, however the challenges identified in table 5.1 present significant barriers to the development and deployment of technical solutions within a law enforcement context. Table 5.1 presents these challenges from a high level, the context within which they manifest in a law enforcement context is explored in more detail in section 5.2.

5.2 Tacit knowledge, algorithms and trust in technology

In their 2020 study which introduced a predictive policing algorithm into an operational policing environment, Ratcliffe et al. (2020) identify a tension between tradecraft and technology. Officers expressed cynicism at technology being able to replicate their experience, and they expressed frustration when the technology did not predict instances of crime. This tension is captured directly in a fieldwork note whereby an officer comments, "*There is a man petting a watering can, and you guys think you can predict where the next burglary is gonna happen?*", (Ratcliffe et al., 2020).

Ratcliffe et al. (2020) ground their analysis of "tradecraft versus technology" within a taxonomy of normative order cultural stuctures proposed by Herbert (1998). Specifically when considering predictive technology, Ratcliffe et al. (2020) sugests that algorithmic policing conflicts against three normative orders, specifically;

1. **Bureaucratic control order**: The concept that an algorithm would define how and where an officer would patrol.
2. **Competence order**: That arbitary deployments via algorithmic prediction would not result in an equal distribution of work across colleagues.
3. **Safety order**: That deployments via algorithmic prediction would not account for the spontaneous need to back up colleagues in danger.

Whilst not grounded in the same normative order theory, observations by Gundhus (2013) suggest that operational policing culture values tacit knowledge and experience over explicit knowledge. Gundhus (2013)'s observations mirror those of Ratcliffe et al. (2020), however Gundhus (2013) grounds her commentary on the cultural differences between occupational professionalism, which is the organic experience and professionalism built by practitioners through experience and autonomy in the workplace, and organisational professionalism, which is the imposed process and practice built by managers through targets and standardisation. Synthesising the work of Ratcliffe et al. (2020) and Gundhus (2013) suggests that the tension between tradecraft and technology may be a tension between tacit occupational professionalism and explicit organisational professionalism. The former values "on the job" knowledge and experience, whereas the latter values explicit knowledge and top down approaches.

Whilst the cultural tension between tacit and explicit experience is called out by Ratcliffe et al. (2020) and Gundhus (2013), Shih (2026) considers diffusion theory in observing the adoption of a technological solution into a policing environment, noting that whilst the uptake of technology by law enforcement may be negatively influenced by those who are staunchly against adoption, a positive driver for adoption is the phenomenon of structural equivalence and the desire to not be seen as lagging behind peers and thereby falling behind in terms of career progression.

Shih (2026)'s observations suggest that the tension between tradecraft and technology may be overcome by leveraging the desire for structural equivalence in practitioners, however, the tension between tacit and explicit knowledge is likely to remain a barrier to adoption of any technical solution which seeks to classify grooming content in an operational law enforcement context. It would therefore be reasonable to suggest that any solution should seek to engage with practitioners in a way which values their experience and tacit knowledge, rather than seeking to replace it with explicit knowledge provided by a technical solution. This is discussed in section 5.3.

Table 5.1: Evidence of Key Challenges to Technology Adoption in Policing

Structural Fragmentation of Policing	**Cultural Resistance to Change**	**Barriers to Technology Adoption**
Policing continues to operate across forty-three police forces plus additional law enforcement agencies. This promotes a culture of duplication, lack of oversight and inconsistent practice, (Halford, 2025; Casey, 2025; Cooper, 2024). This fragmentation prevents the sharing of data and ensures that technologies that are implemented are not interoperable across forces, (Laufs and Borrion, 2022). The absence of national standards governing technical workflows between officers, the Crown Prosecution Service (CPS), and the judiciary compounds this fragmentation (Ng et al., 2025).	Whilst policing practitioners are tecnically capable, (Laufs and Borrion, 2022), policing organisations tend to deploy technology from a "top down" delivery model which is often met with cynicism by practitioners who exhibit frustration at the lack of consideration for technology impact on establised process, (Laufs and Borrion, 2022; Ratcliffe et al., 2020) and who demand tools perceived as police-specific (Halford, 2025). Established working practices and administrative complexity deepen this resistance further (Dubravova et al., 2024).	Laufs and Borrion (2022) identify a lack of financial and political commitment, poor interoperability of systems, and insufficient institutional structures as primary obstacles to technical implementation barriers. This is evidenced starkly in His Majesty's Inspectorate of Constabulary and Fire & Rescue Services (HMICFRS) reports His Majesty's Inspectorate of Constabulary and Fire & Rescue Services (2025, 2026). Ng et al. (2025) find that officers frequently lack training to interpret digital outputs, resulting in missed evidence and weakened investigatons, findings noted by Thompson and Manning (2021); Casey (2025).

5.3 Theoretical frameworks for technology adoption

In 2011, the adoption of mobile data technology by policing was studied by Lindsay et al. (2011) using variants of the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) models. Lindsay et al. (2011) concluded that these theoretical models all failed, noting barriers of low awareness of the benefits of technology, queries around the usability of the equipment, and security and reliability issues. These barriers are reflected by the more recent observations captured in table 5.1, notably by Laufs and Borrion (2022) and Ratcliffe et al. (2020). These barriers are predominantly structural in description, however as occupational structures are built from organisational culture, it is reasonable to suggest that these barriers are a reflection of the unique culture of policing and a manifestation of the observations of Gundhus (2013) and Ratcliffe et al. (2020).

Marciniak (2023) took an alternative approach to considering technology adoption in policing, viewing the deployment of a predictive algorithm for identifying high risk offenders through the a lens of Actor Network Theory (ANT). ANT is described by Dolwick (2009) as a descriptive method of exploring how "actors" (both human and non-human) interact to form networks and, subsequently, how these networks influence the behaviours of the surrounding networks. Whilst indepth consideration of ANT is beyond the scope of this submission, Marciniak (2023) argues that the successful deployment and adoption of a predictive algorithm in policing was, in part, due to the process of "co-construction" whereby officers were obliged to consider algorithmic outputs, but were empowered to scrutinse and, where necessary override thouse outputs when their tacit experience suggested it was necessary. Whilst Marciniak (2023) focusses on one deployment in one Constabulary, the principles of co-construction and engagement with officers in the design and deployment of a technical solution are likely to be relevant across any potential deployment of a grooming classification model in a law enforcement context.

This collaborative process is explored further by Halford (2025) who propose the model of Transformer Led Policing (TLP) in deploying AI solutions into a policing context. TLP is visualised in figure 5.1.

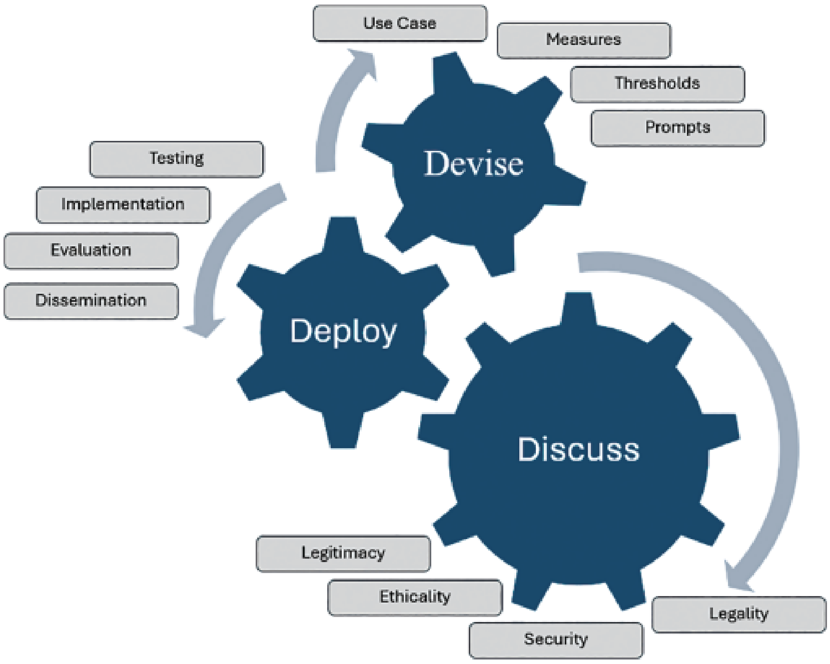

Figure 5.1: Transformer Led Policing model taken from Halford (2025)

Whilst the TLP is noted by Halford (2025) as being applicable to the deployments of generative AI into policing, the notable inclusion of the *discuss* "cog" speaks directly to the findings of Marciniak (2023) that AI innovation within policing needs to be done in collaboration with practitioners, and that the adoption of AI into policing is not a purely technical challenge, but a sociotechnical challenge which requires engagement with the unique culture of policing. This observation circles back to the findings of Gundhus (2013) and Ratcliffe et al. (2020), and suggests that engagement of tacit knowledge in any design will be a critical factor in the successful adoption of any NLP solution into a law enforcement context.

Chapter 6

Ethical and governance considerations

6.1 Core ethical issues in criminal justice

To visualise the thematic scope of ethical considerations in the use of AI within a law enforcement context, a wordcloud from the abstracts from multiple papers is presented in figure 6.1, (Manning and Agnew, 2020; Mwendwa et al., 2026; Oswald, 2021; Babuta et al., 2018; Borgesano et al., 2025; CEPEJ, 2018; Dechesne et al., 2019; Dixon et al., 2021; Erdogan, 2024; Iryna et al., 2025).

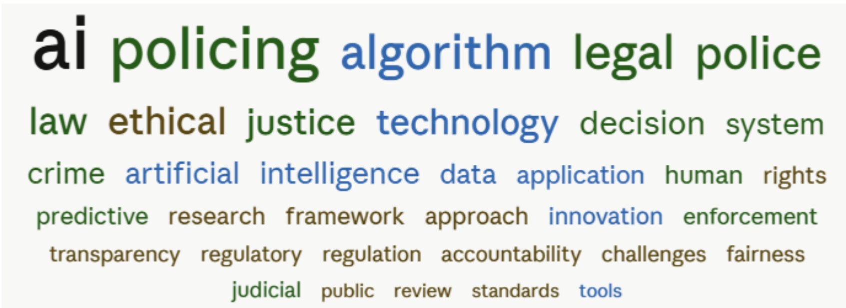

Figure 6.1: Wordcloud of ethical considerations in the use of AI in policing

Figure 6.1 identifies that across recent scholarly papers, ethical considerations into the deployment of AI in policing are spread across multiple domains including legislative, regulatory and technical, in addition to the more obvious themes of accountability and fairness.

Ethical discussions are, by their nature, subjective and context dependent. In an effort to provide a more structured approach to navigating the ethical consideratiions, Babuta et al. (2018) proposes the use of multi-disciplinary ethics boards to provide ethical oversight to the deployment of AI tools in policing, whereas Iryna et al. (2025) suggests that any deployment of AI in policing should be grounded against a staturory framework. Erdogan (2024) takes the suggestion of a statutory obligation further, advocating for an enhanced role for regulatory oversight in the use of AI in policing.

The observations of Babuta et al. (2018), Iryna et al. (2025) and Erdogan (2024) consider the practical considerations of ethical oversight. Manning and Agnew (2020) considers technology in policing from a more abstract sense, noting that the general expansion of AI across society may be rendering the traditional UK model of territorial policing by consent. Reflecting on the commentary in section 5.1, is could e considered that the observations of Manning and Agnew (2020) are manifesting in the operational realities described by Casey (2025), Cooper (2024), Laufs and Borrion (2022) and Ng et al. (2025), resulting in a paradigm whereby the ethical considerations relating to the adoption of AI into policing may need to consider a composite of structural, cultural and technical implementation challenges.

6.2 Practical grounding

Oswald (2021) proposes a three pillar model for ethical oversight of AI in policing, which is visualised in figure 6.2. This model is grounded in practice considerations and parallels the observations of Babuta et al. (2018), Iryna et al. (2025) and Erdogan (2024) in that it considers the practicalities of ethical oversight, rather than the more abstract considerations of Manning and Agnew (2020).

Figure 6.2: Trustworthy use of AI in policing, taken from Oswald (2021)

The three pillar model of Oswald (2021) provides a pragmatic and structured basis for ethical governance that is well-suited to the applied nature of this research. Importantly, Oswald (2021) provides the ethical framework which has been accepted as the foundation for the newly conceived PoliceAI portfolio which is leading the development AI across UK policing, (Jones, 2026).

Rather than engaging with the broader philosophical debates which can result in more theoretical discussions about the nature of ethics in technology, this study adopts Oswald's framework as its primary ethical scaffold.

6.3 Applying ethical principles to the research challenge

The three pillar model of Oswald (2021) provides a practical and structured basis for ethical governance that is well-suited to the applied nature of this research.

Pillar one focusses on the statutory and policy requirements which may govern the use of AI in policing. Within the context of this doctoral journey, statutory instruments such as the Data Protection Impact Assessment (DPIA) and the Data Processing Contract (DPC) provide a mecanism for governance and oversight in the use of law enforcement data in academic research. Marquenie and Quezada-Tavárez (2022) suggests that the authoring of a DPIA needs to consider three factors.

1. Considerations of the controls and safeguards which protect the data.
2. That a value's sensitive design captures requirements ensuring transparancy, accountability and fairness guardrails are implemented.
3. That a there is a human oversight of the data management maintained throughout.

It is notable that whilst the use of a DPIA and a DPC accords with pillar one of Oswald (2021), the considerations of Marquenie and Quezada-Tavárez (2022) simultaneously align with the other pillars, providing the ethical scaffolding a degress of reflective resilience.

The second and third pillars of Oswald (2021) focus on the technical and people aspects of ethical governance. The sequential progression from the DPIA and a DPC submissions into the academic research ethics approval, the technical supervision of the research and the leveraging of the Transformer Led Policing model to engage with practitioners, noted in section 5.3provides a practical and structured basis for ethical governance that is suited to this research. The synthesis of the three pillars of Oswald (2021) with the practical considerations of Marquenie and Quezada-Tavárez (2022) and the practitioner engagement of Halford (2025) is visualised in figure 6.3.

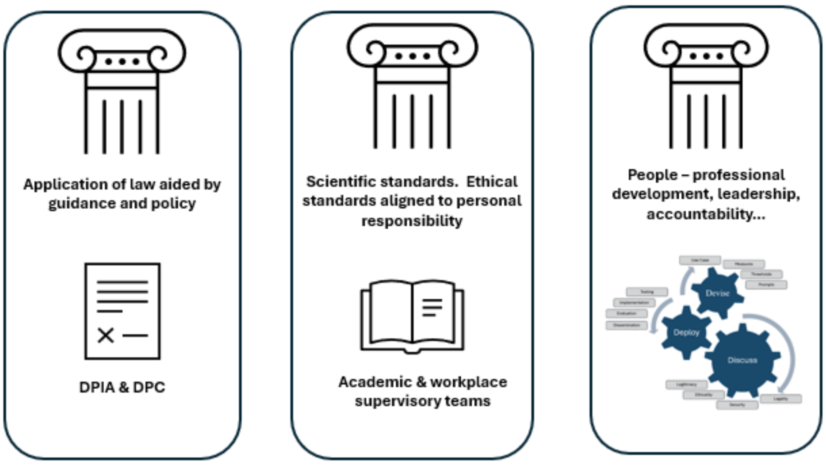

Figure 6.3: Synthesis of Oswald (2021), Marquenie and Quezada-Tavárez (2022) and Halford (2025) to provide a practical and structured basis for ethical governance in this research

Chapter 7

Synthesis, conceptual framework and research questions

7.1 A conceptual framework

7.2 Refined research aim and sub-questions

Chapter 8

Conclusions

8.1 Implications for professional practice

8.2 Next steps

Word count (Chapters 1–N): XXX words, calculated in accordance with standard academic convention, excluding references, figures, tables, captions, running headers, and front matter.

Appendix A

Literature review - search matrix

Table A.1: Literature Review Search Matrix

Analytical Question	Databases	Database Justification
	Track 1 — Criminology	
What linguistic and behavioural precursors of contact sex offending exist in offender–victim conversations, and how do grooming typologies characterise the progression toward contact abuse?	• Scopus • PsycINFO • Web of Science • Criminal Justice Abstracts • ProQuest • Google Scholar	• **Scopus** — Broad peer reviewed coverage across policing and criminology topics. • **PsycINFO** — Consideration of psychological and behavioural dimensions of offending. • **Web of Science** — Broad peer reviewed coverage across policing and criminology topics. • **Criminal Justice Abstracts** — Dedicated criminological literature inclusive of coverage of sex-offending typologies. • **ProQuest** — Includes the Criminal Justice Database focussed on crime-related research, law enforcement practices, and police administration. • **Google Scholar** — Grey literature, NGO reports, and regulatory publications alongside mainstream citation trails.
	Track 2 — Computer Science & NLP	

Continued on next page

Table A continued

Analytical Question	Databases	Database Justification
Which NLP methods can be most effectively realised within policing infrastructure, and what does the evidence base reveal about model performance, explainability, and operational deployment constraints?	• IEEE Xplore • ACM Digital Library • ACL Anthology • arXiv (cs.CL, cs.AI, cs.CY) • Papers With Code • Scopus	• **IEEE Xplore** — Institute of Electrical and Electronics Engineers (IEEE) curated database for applied NLP architecture, development and deployment. • **ACM Digital Library** — Curated by the Association for Computing Machinery (ACM); similar in stature for NLP content as **IEEE Xplore**. • **ACL Anthology** — Curated by the Association for Computational Linguistics (ACL); linguistics-grounded NLP resource with a focus on language understanding and generation, essential for conversational analysis. • **arXiv** (cs.CL) — Pre-print access to novel language model research. • **Papers With Code** — Community driven repository linking code implementations to research papers, crucial for identifying reproducible NLP methods and benchmarks. • **Scopus** — Interdisciplinary bridge between technical and applied criminological literature.
Track 3 — Organisational Structures & Policing		
What structural, cultural, and institutional factors act as barriers to or enablers of NLP adoption in policing organisations, and how does change management theory account for technology resistance in law enforcement?	• Web of Science • Criminal Justice Abstracts • PsycINFO • ProQuest • Dimensions • Google Scholar	• **Web of Science** — Peer-reviewed journals across the social sciences, inclusive of law enforcement and organisational change. • **Criminal Justice Abstracts** — Focused research on policing and policing culture. • **PsycINFO** — Human and organisational literature featuring law enforcement contexts. • **ProQuest** — Includes the Criminal Justice Database focussed on crime-related research, law enforcement practices, and police administration. • **Dimensions** — AI powered search tool for the identification of emerging research clusters in public-sector innovation. • **Google Scholar** — Established open source search engine, used here for grey literature and policy reports from law-enforcement agencies and inspectorates.
Track 4 — Ethics, Law & Governance		

Continued on next page

Table A continued

Analytical Question	Databases	Database Justification
What ethical and regulatory constraints govern NLP deployment in law enforcement, and how should proportionality, data protection, and rights-based frameworks shape the design and oversight of such systems?	• HeinOnline • SSRN • EUR-Lex • PhilPapers • ProQuest & Scopus • Dimensions	• **HeinOnline** — Links to the Law Journal Library containing peer reviewed articles on the ethical implications of AI bias. • **SSRN** — Social Science Research Network (SSRN) contains dedicated sections on criminal justice research focussed on AI and technology adoption. • **EUR-Lex** — Access to European Union (EU) legislative instruments, including the AI Act and GDPR, relevant to algorithmic oversight in law enforcement across the EU. • **PhilPapers** — Repository of academic literature exploring the intersection of ethics, law, and governance regarding police use of AI. • **ProQuest** & **Scopus** — Both offer broad peer reviewed coverage across policing and criminology topics with crossover in ethical and legal research. • **Dimensions** — AI powered search tool for the identification of emerging research clusters in ethical and legal dimensions of AI deployment.

Note. Databases assigned per track based on disciplinary indexing coverage. See Table B for full keyword sets per track.

Appendix B

Keywords and search terms

Table B.1: Search Keywords by Track

Terms	Boolean Strings
Track 1 — Criminology	

• Online grooming
• Sexual grooming
• Grooming behaviour
• Child sexual abuse
• CSA
• CSAM
• Grooming language
• Grooming communication
• Child
• Victim
• Sex offend*
• Abuse
• Contact offending
	`("online grooming" OR "sexual grooming" OR "grooming behaviour") AND ("child sexual abuse" OR "CSA" OR "CSAM")`
	
	`("grooming language" OR "grooming communication") AND ("child" OR "victim") AND ("sex offend*" OR "abuse")`
	
	Proximity (Scopus / WoS):
	`"grooming" NEAR/5 "contact offending"`

Track 2 — Computer Science & NLP	

Continued on next page

Table B continued

Terms	Boolean Strings
• BERT • RoBERTa • DistilBERT • Text classification • Sequence classification • Grooming • Predatory • Child safety	("BERT" OR "RoBERTa" OR "DistilBERT") AND ("text classification" OR "sequence classification") AND ("grooming" OR "predatory" OR "child safety")
• Explainable AI • XAI • Interpretable machine learning • NLP • Natural language processing • Law enforcement • Policing • Criminal justice	("explainable AI" OR "XAI" OR "interpretable machine learning") AND ("NLP" OR "natural language processing") AND ("law enforcement" OR "policing" OR "criminal justice")
• PAN shared dataset • Perverted Justice • Chat log • Online conversation • Grooming detection • Predatory communication • Sexual solicitation	("PAN shared dataset" OR "Perverted Justice" OR "chat log" OR "online conversation") AND ("grooming detection" OR "predatory communication" OR "sexual solicitation")
Track 3 — Organisational Studies & Policing	
• Technology adoption • Technology resistance • Organisational inertia • Policing • Law enforcement • Police force	("technology adoption" OR "technology resistance" OR "organisational inertia") AND ("policing" OR "law enforcement" OR "police force")
• Tacit knowledge • Explicit knowledge • Knowledge management • Police officer* • Police culture • Innovation	("tacit knowledge" OR "explicit knowledge" OR "knowledge management") AND ("policing" OR "police officer*" OR "law enforcement") *Proximity (WoS):* "police culture" NEAR/8 ("technology resistance" OR "innovation")
Track 4 — Ethics, Law & Governance	

Continued on next page

Table B continued

Terms	Boolean Strings
• Algorithmic bias • Automated decision-making • Algorithmic accountability • Law enforcement • Policing • Criminal justice • Ethics • Human rights • Fairness	`("algorithmic bias" OR "automated decision-making" OR "algorithmic accountability") AND ("law enforcement" OR "policing" OR "criminal justice") AND ("ethics" OR "human rights" OR "fairness")`
• Data protection • GDPR • UK DPA • NLP • Machine learning • Artificial intelligence	`("data protection" OR "GDPR" OR "UK DPA") AND ("NLP" OR "machine learning" OR "artificial intelligence") AND ("policing" OR "criminal justice")`
Cross-Track — Consolidated searches	
• Natural language processing • NLP • Text classification • Grooming • Child sexual exploitation • Online predator • Detection • Identification • Classification	`("natural language processing" OR "NLP" OR "text classification") AND ("grooming" OR "child sexual exploitation" OR "online predator") AND ("detection" OR "identification" OR "classification")`
• Machine learning • Artificial intelligence • Policing • Law enforcement • Implementation • Adoption • Deployment • Barrier* • Challenge* • Constraint*	`("NLP" OR "machine learning" OR "artificial intelligence") AND ("policing" OR "law enforcement") AND ("implementation" OR "adoption" OR "deployment") AND ("barrier*" OR "challenge*" OR "constraint*")`

Appendix C

Inclusion criteria

Table C.1: Inclusion Criteria

Inclusion Criteria

IC1	Peer-reviewed articles, conference proceedings, and edited book chapters, 2018–present.
IC2	Grey literature from authoritative sources (National Police Chiefs' Council (NPCC), Home Office, Child Exploitation and Online Protection (CEOP), National Institute of Standards and Technology (NIST), Information Commissioner's Office (ICO)), 2018–present.
IC3	NLP, machine learning, or computational linguistics applied to safeguarding, criminal justice, or analogous operational contexts.
IC4	Grooming, CSA, or online predatory behaviour with linguistic, behavioural, or empirical focus.
IC5	Organisational, cultural, or governance studies on technology adoption in policing or structurally comparable public-sector contexts.
IC6	English-language sources; non-English permitted where a peer-reviewed translation or substantive abstract provides sufficient evidential grounding with explicit methodological justification.
IC7	Pre-2018 foundational works where theoretical significance is established and no updated equivalent exists, subject to explicit justification in the synthesis narrative.

Appendix D

Exclusion criteria

Table D.1: Exclusion Criteria

Exclusion Criteria	
EC1	Sources prior to 2018 unless meeting IC7; all exceptions recorded in the review log.
EC2	Non-peer-reviewed opinion pieces, blog posts, or media articles without evidential grounding, unless from authoritative professional or statutory bodies.
EC3	Studies focused solely on adult sexual offending with no transferable relevance to child safeguarding contexts.
EC4	Technical NLP literature without contextual application to safeguarding, criminal justice, or operational deployment (foundational instructional / code text not included)
EC5	Technology adoption studies from sectors lacking structural or cultural comparability to UK policing (e.g. private-sector contexts without analogous accountability frameworks).
EC6	Duplicate publications; where multiple versions exist, the most recent peer-reviewed version is retained.
EC7	Sources for which full text cannot be obtained via institutional access, inter-library loan, or open-access repositories within a reasonable retrieval window.

Appendix E

Patch One - Upload and Feedback

Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four literature tracks to develop a conceptual framework capable of addressing the proposed research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

This challenge consists of four thematic research areas:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
2. Which NLP methods can be most effectively realised within policing infrastructure?
3. What implementation barriers exist within policing that may hinder NLP adoption?

4. What strategies can be developed to overcome those barriers?

1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. The four tracks are as follows:

1. The first track explores the landscape of CSA and the linguistic characteristics which precede contact offending, establishing the empirical foundation for a technical solution.
2. The second track reviews the technical literature on NLP applications in safeguarding and criminal justice contexts, focusing on grooming language detection and the computational methods available for identifying linguistic precursors of contact offending.
3. The third track examines the organisational dynamics of technology adoption in policing, drawing on both academic literature and empirical evidence of structural and cultural barriers to technical innovation.
4. The fourth track addresses the ethical and governance boundaries within which any NLP solution must operate, encompassing algorithmic bias, data protection, proportionality, and the ethical context surrounding preventative safeguarding.

Each track is explored in depth, and their insights are then synthesised to identify where technical possibilities meet organisational realities. This structure enables a comprehensive understanding of both the capabilities and constraints shaping both the research problem and the practice improvement opportunity and grounding any solution in organisational and ethical realities

Chapter 2

Literature Search Strategy and Methodology

2.1 Review Methodology and Rationale

This literature review adopts a systematic integrative approach. Snyder (2019) documents a number of review types, noting that an integrative review seeks to combine multiple perspectives, synthesise those views, and to develop a new framework or perspective.

The interdisciplinary nature of this research, noted in section 1.2, necessitates a review methodology that can accommodate multiple perspectives and evidence types. An integrative design as defined by Torraco (2016) and Torraco (2005) allows for a review and synthesis across domains, building a conceptual framework which seeks to unify the disparate domains into a framework for operationalising and implementing an NLP solution into a safeguarding CSA context.

A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identifies that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a layered systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demands a robust literature and note management strategy supported by appropriate tools. Basu (2020) describes the Zettelkasten (ZK) method which can be used to manage literature sources and create links between them, supporting the synthesis process. This is furthered by Reck (2023) who describes the ZK method as a process of creating notes on individual literature sources, applying tags to those notes and then using the tagging system to create links between the notes. Both Basu (2020) and Reck (2023) describe the ZK method as the process of building a Personal Knowledge Management System (PKMS). Cheong and Tsui (2011) describes a PKMS as a structure which allows for the development of "*personal information management, personal knowledge internalization, personal wisdom creation, and interpersonal knowledge transferring*"

Reck (2023) considers the use of various software tools to support the ZK method. Following a period of experimentation with various applications, a software stack to support the ZK method for this review was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note taking and synthesis and LaTeX for writing and formatting the review.

Zotero was selected for its robust literature management capabilities, including the ability to capture metadata, PDFs and notes on literature sources, and to auto generate and update a master bibliography file. Obsidian was selected for its flexibility in note taking and synthesis, allowing for the creation of a network of linked notes which can be easily navigated and searched. LaTeX was selected for its professional formatting capabilities, particularly for handling citations, figures and tables, and for its ability to draw directly from Zotero's auto-generated bibliography, thus ensuring consistency and ease of referencing.

All content is then automatically synced to a cloud store and defined content saved a second time to a GitHub repository. My use of the ZK method and the associated software stack is illustrated in figure 2.1.

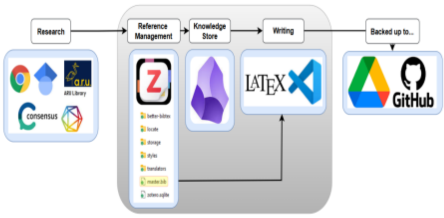

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

Section 1.2 identifies four distinct research tracks. Each of these tracks has a unique focus and therefore requires a dedicated search strategy to maximise the relevance and quality of included literature. The search strategy presented in table ?? defines the four core research tracks, the question to be answered for each track, the selected databases for each track, and the rationale for selecting those databases. The search strategy is designed to ensure that the literature review is comprehensive, targeted and relevant to the research challenge and that it can be replicated by other researchers. The search strategy is also designed to ensure that the literature review is manageable, given the breadth of the research challenge and the interdisciplinary nature of the review.

2.4 Keywords and search terms

Key words and search terms are critical to the success of a literature review. Chigbu et al. (2023) provides holistic guidance on keyword selection and submission to selected databases. The keywords and search terms utilised for each of the noted research tracks are defined in table ??. These align to the search strategy outlined in table ??. The keywords and search terms are designed to capture the key concepts and themes relevant to each track, ensuring that the literature retrieved is relevant to the research challenge.

aru

Patrick Thompson

18 Mar 20:18 | Last reply 3 Apr 23:42

Hi squad!

I've uploaded a PDF entitled "Draft so far..." which is an effort to get some structure around my literature review.

I've not made much headway on actually reading lots yet as I wanted to nail down the strategy.

Hopefully the document makes logical sense. Essentially I've broken down my literature review into four tracks / topics and then decided on a systematic integrative review methodology given the four slightly separate but linked topic areas.

In the files section (https://canvas.anglia.ac.uk/groups/44479/files) I've also uploaded four tables;

One is called search_matrix. Its a table were I try to justify search sources for the four tracks.

One is called keywords. It's my attempt to narrow down keywords across those four tracks.

The other two are inclusion / exclusion criteria. The 2018 date is linked to the year when large language models became more openly available and made a big difference in the world of natural lanuage processsing.

Speak soon!

Pat

Name ▴	Created	Last Modified	Modified by	Size	Actions
< > exclusion_criteria.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	2 KB	⋮
< > inclusion_criteria.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	2 KB	⋮
< > keywords_v2.html	25 Mar 2026	25 Mar 2026	Patrick Thompson	17 KB	⋮
< > search_matrix.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	8 KB	⋮

Beth Brown

29 Mar 14:57

Hi Pat,

For an early first draft I thought this was awesome! Your writing is clear and you articulate the research context with precision, I felt that your opening section clearly situates the problem within policing, policy, and technological landscapes, and you explain this with a balanced, evidence-based tone.

The four research tracks are logical and well reasoned, clearly connecting to your overarching question. Your tables are also incredibly clear and organised. I think the fact you have got such a clear structure will make your literature review manageable and robust. It also shows that you've thought deeply about the complexity of the problem space.

The justification for separate strategies is sound and aligns well with the assignment brief. You could consider briefly mentioning in the introduction what is out of scope and why. This may help manage expectations and demonstrate deliberate boundary setting. Also, you've clearly explained the tracks individually but a short statement about how they will be brought together (e.g., thematic synthesis, conceptual mapping, cross-domain triangulation) would provide further clarity about the approach.

I really liked your explanation of why an integrative review is appropriate, and how you're layering systematic elements to ensure rigour. This demonstrates a strong understanding of review methodologies and defended your choices well.

Overall I thought this was great - you show really clear thinking and a strong grasp of both the technical and organisational sides of the topic! Look forward to the next Patch..

Reply | Like | Mark as unread

Lauren Hunt

3 Apr 23:42

Hi Pat,

Its reassuring we're at a similar stage, but its clear you know your topic and that you've spent a lot of time thinking of the best way to tackle it!

Love the tables in the files and numbering your in/exclusion criteria to refer to. You (and Beth) have showed your Boolean search words very well and something I'd develop in my work - I also like how you clearly showed each element and then showed the cross of the topics (even if I don't understand some of the AI bits!). Its good to see how all of your tracks meet up and relate to your question - you've done well to get into the detail of the topic but also explain each section clearly.

The search matrix took me a second to get my head around, but this only helped me as again you and Beth have used many more databases/journals than me so maybe I need to explore this more to maximise results. I was worried about over-complicating my search but you've both presented this in clear formats so I may give it another go! I can also see how your feedback to me to clear in your own work regarding in/exclusion of paper dates, with 2018 being an important time for your topic.

You justify your choices well with citations and your writing flows nicely - overall I think you've made a fantastic start and time well spent setting up a great foundation - I look forward to draft 2!

Reply | Like | Mark as unread

Patrick Thompson

24 Mar 16:11

Hi Beth,

Here are my thoughts!

The keyword choices are well considered and the reflective bit around spotting "constabulary" in the results shows non-linear thinking... you're engaging with the process rather than just running through it mechanically. The Boolean searches you settled on clearly gave you a really solid foundation of papers to work from.

My dodgy maths puts you somewhere around 5,700 returns once you focus down to policing – though with three databases there's inevitably going to be a hefty overlap, so I'm guessing the real number is probably closer to 2K.... is there scope to be more robust on the exclusion criteria? You've only got one at the moment and I wonder if tightening this would actually give you more freedom, letting you really focus on contemporary practice and lived officer experience rather than having to wade through how things have evolved over the last decade and a half?

I guess along side that is the 2010 date. Sixteen years is a long stretch and I wonder if it's worth either shortening the time span or giving that choice a really solid justification, especially since the five-year Google Scholar search felt a little inconsistent alongside it.

I really liked the thematic buckets in the screening section – they're well chosen and make sense. It did make me wonder though how you're moving from the initial screening to identifying *key* papers specifically? It might be worth a sentence on what's elevating one paper above another?

The critical summaries are cool!

Reply | Like | Mark as unread

Patrick Thompson

24 Mar 16:38

Hi Lauren,

I really like the way you've set this out – the PICO structure to define the requirements is really logical, and the dual-track in-vivo and in-vitro approach makes the methodology robust.

The inclusion and exclusion criteria are sensible and clearly reasoned. One question worth considering: does there need to be any date parameter in the search, or at least an explicit justification for why all dates are included? Reviewers may ask about this.

The planned PRISMA flow chart is a great addition and will keep the whole process transparent and reproducible. Looking forward to seeing the work develop – you seem to be at a similar stage to me in having nailed your methodology and are about to start the actual capture of content.

Reply | 1 Like | Mark as unread

Appendix F

Patch Two - Upload and Feedback

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